

1.6 - Expansion of the Wakes

In practice we are mostly interested in knowing the wake functions in a region very close to the vacuum chamber's center - where the beam travels. In that region $\vec{\rho}_s$ and $\vec{\rho}_w$ are much smaller than the transverse dimensions of the chamber, allowing us to expand W in its transverse coordinates and keep only low-order terms:

$$\begin{aligned} W(\vec{\rho}_s, \vec{\rho}_w, z) &\approx W_0 + \vec{\rho}_s^T \cdot \vec{M}_s + \vec{\rho}_w^T \cdot \left(\vec{M}_w + D \cdot \vec{\rho}_s + \frac{1}{2} Q \cdot \vec{\rho}_w \right) = \\ &= W_0 + \underbrace{\vec{\rho}_s^T \cdot \vec{M}_s + \vec{\rho}_w^T \cdot \vec{M}_w}_{\text{monopole terms}} + \underbrace{\vec{\rho}_w^T \cdot D \cdot \vec{\rho}_s}_{\text{Dipole term}} + \underbrace{\frac{1}{2} \vec{\rho}_w^T \cdot Q \cdot \vec{\rho}_w}_{\text{Quadrupole term}} + \frac{1}{2} \vec{\rho}_s^T \cdot Q \cdot \vec{\rho}_s + \dots \end{aligned}$$

↳ less relevant

where W_0 , M_s , M_w , D and Q are dependent on z

Recovering the wake functions (until 1st order on $\vec{\rho}_s$ and $\vec{\rho}_w$)

$$\bullet \quad w_{||} = \frac{\partial W}{\partial z} \approx \underbrace{W_0}_{\text{usually the only term considered for } w_{||}} + \vec{\rho}_s^T \cdot \vec{M}_s + \vec{\rho}_w^T \cdot \vec{M}_w + \dots$$

* usually the only term considered for $w_{||}$

dominates the expansion ~ usually referred to as the longitudinal wake
→ responsible for longitudinal effects:

- ▼ avg. energy loss
- ▼ potential-well distortion
- ▼ bunch-lengthening/shortening
- ▼ coupled bunch oscillations

$$\bullet \quad \vec{w}_\perp = \vec{\nabla}_{w\perp} W \approx \vec{M}_w + D \cdot \vec{\rho}_s + Q \cdot \vec{\rho}_w$$

* $\vec{M}_s$ and $\vec{M}_w$ ~ usually referred to as monopolar wakes → effect independent of the transv. displacement of the source and wita. particles
→ change in the closed orbit ~ does not lead to instabilities
as a func. of the current

→ in most practical cases these terms are zero due to symmetries in the vacuum chamber

★ $\mathcal{D} = \begin{pmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{pmatrix} \sim D_{ij}'$ are referred to as dipolar wakes ~ as they are originated from an offset of the source particle from the chamber's center, and

↳ $D_{11} \sim$ horizontal dipolar wake

↳ $D_{22} \sim$ vertical dipolar wake

Generates a wake:

$$\vec{w}_1^D = \mathcal{D} \cdot \vec{p}_s, \text{ which is thus independent from } \vec{p}_w,$$

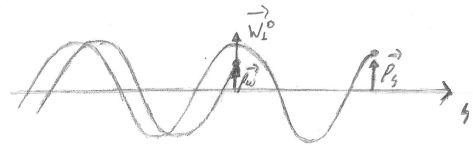
Just like in the case of a mag. dipole

generate a wake that is "dipole-like", in the sense that it doesn't depend on the coord.'s x and y of the witness particle!

↳ Responsible for the transverse oscillations that cause coupled bunch oscillations, emittance deterioration and even beam loss

* These terms are particularly dangerous as they describe a mechanism where oscillations of the source particle can induce oscillations on the witness particle, which always leads to a resonant effect

as both particles naturally already oscillate with very similar frequencies (equiv. to saying they have approx. the same tune)



→ $D_{12} = D_{21}$ is usually zero due to symmetries in the vacuum chamber, and even when it is not, its effects are not significant as the hor. and vert. tunes are very different, and thus no resonant effects appear - oscillations in one plane do not transfer to oscillations in the other.

★ Q_{11} and $Q_{22} \sim$ Detuning or Quadrupolar wakes

$\vec{w}_1^Q = Q \cdot \vec{p}_w \rightarrow$ Wake/kick whose strength depends on the transv. coordinates of the witness particle itself $\rightarrow$ just like in a quadrupole magnet

Furthermore, if $v \approx c$,
 $Q_{22} = -Q_{11}$, producing opposite focalizations on x and y , precisely as the mag. quad. does

$\rightarrow$ generates a focusing effect

$\sim Q_{12} = Q_{21}$ usually = 0 from symmetry considerations

Further understanding the symmetry implications and simplifications that become evident from the wake potential $W(\vec{\rho}_s, \vec{\rho}_w, z)$ formalism (highlighting its power):

Again,

$$W(\vec{\rho}_s, \vec{\rho}_w, z) = W_0 + \vec{\rho}_s^T \cdot \vec{M}_s + \vec{\rho}_w^T \cdot \vec{M}_w + \vec{\rho}_w^T \cdot \mathcal{D} \cdot \vec{\rho}_s + \frac{1}{2} \vec{\rho}_w^T \cdot \mathcal{Q} \cdot \vec{\rho}_w$$

Now, let's consider a chamber with mirror symmetry (e.g. rectangular or elliptical)
($x \rightarrow -x$ and/or $y \rightarrow -y$ do not change W)

Take for instance $x \rightarrow -x$: (exception example: radiation masks)

$$W(-x_s, y_s, -x_w, y_w, z) = W(+x_s, y_s, +x_w, y_w, z)$$

$$\bullet (-x_s \ y_s) \cdot \begin{pmatrix} M_{sx} \\ M_{sy} \end{pmatrix} = (x_s \ y_s) \cdot \begin{pmatrix} M_{sx} \\ M_{sy} \end{pmatrix} \Rightarrow -M_{sx} = M_{sx} \Rightarrow M_{sx} = 0$$

• Analogously for $\vec{\rho}_w^T \cdot \vec{M}_w$, we get $M_{wx} = 0$

† Doing the same but with $y \rightarrow -y$, we get: $M_{sy} = 0$, $M_{wy} = 0$ *

$$\bullet (-x_w \ y_w) \cdot \begin{pmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{pmatrix} \cdot \begin{pmatrix} -x_s \\ y_s \end{pmatrix} = (-x_w \ y_w) \cdot \begin{pmatrix} -D_{11}x_s + D_{12}y_s \\ -D_{21}x_s + D_{22}y_s \end{pmatrix} = D_{11}x_w x_s - D_{12}x_w y_s - D_{21}x_s y_w + D_{22}y_s y_w,$$

$$\text{while } (x_w \ y_w) \cdot \mathcal{D} \cdot \begin{pmatrix} x_s \\ y_s \end{pmatrix} = D_{11}x_s x_w + D_{12}x_s y_w + D_{21}x_w y_s + D_{22}y_s y_w,$$

and thus, their equality for any x_s, y_s, x_w, y_w demands:
$$\begin{cases} -D_{12} = D_{12} \Rightarrow D_{12} = 0 \\ -D_{21} = D_{21} \Rightarrow D_{21} = 0 \end{cases}$$

• Proceeding analogously for $\vec{\rho}_w^T \cdot \mathcal{Q} \cdot \vec{\rho}_w$, we find $Q_{12} = Q_{21} = 0$

$\Rightarrow$ Non-diagonal terms of $\mathcal{D}$ and $\mathcal{Q}$ are all zero!

Furthermore, since W is an harmonic function on x_w and y_w for ultra-relativistic electrons,

$$\nabla_{w\perp}^2 W = \frac{\partial^2 W}{\partial x_w^2} + \frac{\partial^2 W}{\partial y_w^2} = 0, \text{ and the quadratic terms with } x_w \text{ and } y_w \text{ on the } W$$

expansion (which can thus "survive" two "witness-transverse" derivatives) are:

$$2 \cdot W^{\text{quad}} = \vec{\rho}_w^T \cdot Q \cdot \vec{\rho}_w = Q_{11} x_w^2 + 2 Q_{12} x_w y_w + Q_{22} y_w^2, \text{ we have:}$$

$$\frac{\partial^2 W}{\partial x_w^2} + \frac{\partial^2 W}{\partial y_w^2} = Q_{11} + Q_{22} = 0 \Rightarrow \boxed{Q_{11} = -Q_{22}}$$

And thus for a vacuum chamber w/ mirror symmetry containing ultra-rel electrons:

$$\begin{aligned} \rightarrow \vec{M}_b = \vec{M}_w = \vec{0} \quad & \rightarrow D = \begin{pmatrix} D_{11} & 0 \\ 0 & D_{22} \end{pmatrix}; \quad \rightarrow Q = \begin{pmatrix} -Q_{22} & 0 \\ 0 & Q_{22} \end{pmatrix} \end{aligned}$$

and W 's expansion simplifies to:

$$W(\vec{\rho}_s, \vec{\rho}_w, z) = W_0 + D_{11} x_s x_s + D_{22} y_w y_s - \frac{1}{2} Q_{22} x_w^2 + \frac{1}{2} Q_{22} y_w^2 //$$

W_0, D_{11}, D_{22} and Q_{22} all functions of z .

This enables recovering the recurrent wake functions found in the literature:

defining $W_x^D \triangleq D_{11}$, $W_y^D = D_{22}$, and $W^Q = Q_{22}$, we get:

$$\left\{ \begin{aligned} \bullet W_{11}(z) &= \frac{\partial W}{\partial z} = W_0'(z) \\ \bullet W_x(x_s, x_w, z) &= W_x^D(z) \cdot x_s - W^Q(z) \cdot x_w \\ \bullet W_y(y_s, y_w, z) &= W_y^D(z) \cdot y_s + W^Q(z) \cdot y_w \end{aligned} \right. //$$

To determine such terms, one can first do a simulation with $x_s \neq 0, x_w = 0$ to find $W_x^D (= w_x/x_s)$, and then perform another sim. w/ $x_s = 0, x_w \neq 0$ to get $W^Q (= -w_x/x_w)$!

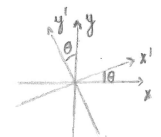
Lastly, if the vacuum chamber presents not only mirror symmetry, but also

rotation symmetry — i.e., cylindrical symmetry — W must also satisfy:

(exception example: undulators)

$$W(R \cdot \vec{\rho}_s, R \cdot \vec{\rho}_w, z) = W(\vec{\rho}_s, \vec{\rho}_w, z),$$

where R is a rotation matrix $\left(\begin{smallmatrix} \oplus \\ R = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \end{smallmatrix} \right)$



$$\begin{aligned} x' &= x \cos \theta + y \sin \theta \\ y' &= -x \sin \theta + y \cos \theta \\ \Rightarrow \begin{pmatrix} x' \\ y' \end{pmatrix} &= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \end{aligned}$$

Plugging into the wake expansion:

$$W(R \cdot \vec{\rho}_s, R \cdot \vec{\rho}_w, z) = W_0 + \vec{\rho}_w^T (R^T \cdot D \cdot R) \vec{\rho}_s + \vec{\rho}_w^T (R^T \cdot Q \cdot R) \vec{\rho}_w$$

And thus we need:

$$\begin{aligned} \bullet R^T \cdot D \cdot R &= D \Rightarrow \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} D_{11} & 0 \\ 0 & D_{22} \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} = \begin{pmatrix} D_{11} & 0 \\ 0 & D_{22} \end{pmatrix} \\ &= \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} D_{11} \cos \theta & D_{11} \sin \theta \\ -D_{22} \sin \theta & D_{22} \cos \theta \end{pmatrix} = \begin{pmatrix} D_{11} \cos^2 \theta + D_{22} \sin^2 \theta & (D_{11} - D_{22}) \sin \theta \cos \theta \\ (D_{11} - D_{22}) \sin \theta \cos \theta & D_{11} \sin^2 \theta + D_{22} \cos^2 \theta \end{pmatrix} = \end{aligned}$$

which can only be true if: $D_{11} = D_{22}$, i.e., $W_x^D = W_y^D = W^D$

• Analogously, $R^T \cdot Q \cdot R = Q \Rightarrow Q_{11} = Q_{22}$, but from $\nabla_{\perp}^2 W = 0$ we have $Q_{11} = -Q_{22}$,

leaving as the only option: $Q_{11} = Q_{22} = 0 \Rightarrow W^Q = 0 \Rightarrow$ No quadrupolar wake!

$\Rightarrow$ Therefore, for a perfectly round chamber, $\vec{M}_s = \vec{M}_w = 0$; $D = \begin{pmatrix} W^D & 0 \\ 0 & W^D \end{pmatrix} = W^D \cdot \mathbb{I}$; $Q = 0$, and

the wakes become:

$$\begin{aligned} \bullet W_H(z) &= W_0(z) \\ \bullet W_x(x_s, z) &= W^D \cdot x_s \\ \bullet W_y(y_s, z) &= W^D \cdot y_s \end{aligned} \left\{ \vec{W}_{\perp} = W^D \cdot \vec{\rho}_s \right.$$

Thus, in short:

◆ General case:

$$\bullet w_{||}(z) = W'_0(z)$$

$$\bullet \vec{w}_\perp(\vec{\rho}_s, \vec{\rho}_w, z) = \vec{M}_w(z) + D \cdot \vec{\rho}_s + Q \cdot \vec{\rho}_w$$

◆ Mirror-symmetry

$$\bullet w_{||}(z) = W'_0(z)$$

$$\bullet w_x(x_s, x_w, z) = W_x^D(z) \cdot x_s - W_x^Q(z) x_w$$

$$\bullet w_y(y_s, y_w, z) = W_y^D(z) \cdot y_s + W_y^Q(z) y_w$$

◆ Axisymmetry

$$\bullet w_{||}(z) = W'_0(z)$$

$$\bullet w_x(x_s, z) = W^D(z) \cdot x_s$$

$$\bullet w_y(y_s, z) = W^D(z) \cdot y_s$$